Dušan Blanuša

Electrical and computer science engineer

Personal information:

Website:
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Date and place of birth:
19.01.1998. Kragujevac, Serbia

Residence:
Novi Sad, Serbia

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Technologies:

Programming Languages: Python, Java, JS, Angular, React

AI Technologies: LLMs, Evolutionary Algorithms, OpenCV, Keras, Tensorflow, NLTK, NLP, OCR, Spacy, Scikit-learn, Gensim, Yolo, Keras, Pandas, SARIMA

Database: Oracle, MySQL, PostgreSQL, MongoDB, Neo4j, Elastic, CosmosDB

IDEs: Visual Studio, PyCharm, VSCode

Methodologies: Agile, Scrum
Project Management Tools: JIRA, Git, Miro

Operating Systems: Linux, Windows

Education

Bachelor of Electrical and Computer Engineering
Faculty of Technical Sciences, University of Novi Sad
Pursuing master's degree in electrical and Computer Engineering
Faculty of Technical Sciences, University of Novi Sad

TRAINING/ CERTIFICATIONS
Synechron Next Gen Leadership
Azure AI 900
NLP Training – Synechron

Synopsis

- Technology Lead with 7 years of experience in Artificial Intelligence and Data Science, delivering production-grade AI solutions
- Specialized in LLM systems, multi-agent architectures, and distributed ML platforms
- Led development of complex AI systems, including RAG pipelines, memory-enabled agents, and large-scale data processing solutions
- Experienced in designing scalable architectures, leading cross-functional teams, and driving projects from concept to production
- Strong focus on system design, performance optimization, and business impact of AI solutions
- Proactive and results-driven professional with strong leadership, communication, and mentoring skills

Experience

- Technology Lead | Synechron
  (Apr' 2021 to present)

  - Developing modern AI solutions from initial ideas and creating POCs and MVPs, through designing architecture, implementation, monitoring team progress, organizing tasks, to presenting final solutions to client executives.
  - Leading web teams and collaboration with them in the implementation of complete solutions
  - Communication with clients, understanding their issues, and negotiating workflow in line with their requirements and feasible objectives.
  - Team leadership and resource organization.
  - Conducting AI trainings for juniors.
  - Conducting interviews for new hires.
  - Delivering lectures on the applications of modern AI solutions, predominantly LLMs
  - Presenting the company and delivering lectures about it.

- Intern | Synechron
  (Sep' 2020 to Nov' 2020)

  - The internship included work on projects based on NLP systems and recommendation systems using the Python programming language.
  - Involved in Python programming and creating AI models.
  - Responsible for data research and data analysis.

- Programmer | Memristor Robotics
  (Feb' 2019 to Aug' 2020)

  - Worked part-time on creating computer vision solutions for the robots.
  - Utilized the Python programming language with various libraries for computer vision and artificial intelligence, across Linux operating systems.
  - Team leadership and resource organization, from creating datasets to the final implementation.

- Teaching assistant | Faculty of Technical Sciences
  (Oct' 2018 to Jul' 2019)

  - Teaching assistant at the Department of Computer Technology and Computer Communications.
  - Teaching assistant at the Department of Applied Computer Science and Informatics.

- Editor| emobilnost.rs
  (Jan' 2018 to Oct' 2018)

  - Worked on articles about autonomous vehicles, systems for their control, and the safety of modern autonomous vehicle systems.
  - Maintained a website.

Projects

- **Browser Agent| Synechron | Technology Lead | Team size: 6**
  Responsibilities: Designing and developing browser automation agent using LLMs and deep agents. Automating browser-based workflows for data collection and research. Integrating vision-based models for interacting with web interfaces.
  Enviropment: Python, Playwright, LangChain, Vision LLMs

- **Document service| Synechron | Technology Lead | Team size: 6**
  Responsibilities: Designing and developing document service for generating documents with citations from multiple sources. Implementing authentication and authorization mechanisms for secure document access. Ensuring traceability and reliability of AI-generated outputs.
  Enviropment: Python, FastAPI, Microsoft authentication, AWS databases

- **Text to SQL and Router agent| Synechron | Technology Lead | Team size: 4**
  Responsibilities: Implementing memories for the existing text to SQL agent. Prototyping, designing and implementing router agent for multy agents system orchestration based on the intent classification. Implementing fast api background.
  Enviropment: Python, LlamaIndex, Mem0, AWS, FastAPI, Redis, Qdrant, OpenSearch, Bitbucket

- **Deep analysis of XML risk documents | Synechron | Artificial Intelligence engineer | Team size: 7**
  Responsibilities: Creation of a custom parser for narrow-domain documents in XML format, system for linking logical parts of different documents and RAG system on top of that. Developing regression models for predicting risks for public companies. Leading frontent React team.
  Enviropment: Python, Neo4j, CosmosDB, Azure AI search, OpenAI, Langchain, Databricks, Scikit-learn, React, Azure DevOps

- **Chatbot based on RAG system | Synechron |Technology Lead | Team size: 5**
  Responsibilities: Planning project and creating Miro board graphs. Designing and coding project architecture, databases and chatbot intents and dialogs. Leveraging LLMs with Azure Bot Framework for parsing unstructured PDF documents.
  Enviropment: Python, Azure Bot Framework, MongoDB, CosmosDB, Elastic search, Azure AI search, GIT

- **Technical Insights Generator | Synechron | Technology Lead | Team size: 4**
  Responsibilities: Planning project. Designing and coding project architecture and databases. Organizing team and programming tasks. Leading frontent React team. Extracting usefull data from web-scraped documents. Creating topic modeling, trend monitoring, summarization and report-creating systems.
  Enviropment: Python, React, MongoDB, Neo4j, GPT, FastAPI, GIT

- **Intelligent retirement adviser | Synechron | Artificial Intelligence Engineer | Team size: 2**
  Responsibilities: Planning and creating systems for personalizing retirement plans, calculating ideal retirement plan for desired goal, calculating pension payments and duration, customized economics interest and loan formulas. Creating intelligent retirement planning adviser based on evolutionary algorithms and custom fitness functions
  Enviropment: Python, Evolutionary Algorithms, SARIMA, FastAPI, GIT

- **FAQ LLM Chatbot | Synechron | Artificial Intelligence Engineer | Team size: 4**
  Responsibilities: Making FAQ chatbot for corporate questions. Creating systems for collecting company data and for question answering based on provided context, leveraging LLMs
  Enviropment: Python, MongoDB, GPT, GIT

- **LLM coding assistant | Synechron | Artificial Intelligence Engineer | Team size: 4**
  Responsibilities: Leveraging different LLMs in coding assistant system, prompt engineering.
  Enviropment: Python, LLMs, MongoDB, GIT

- **Involvement predictions | Synechron | Artificial Intelligence Engineer | Team size: 6**
  Responsibilities: Automating the manual task of classifying companies on over 40 classes based on collected data and creating intelligent systems to draw conclusions and streamline the work for data analysts. Communicated with four different teams to understand their needs and integrated their ideas into a successful solution that is now in use.
  Enviropment: Python, MongoDB, Flask, NLTK, GIT

- **Phone damage claims based on computer vision| Synechron | Artificial Intelligence Engineer | Team size: 1**
  Responsibilities: Making system for phone damage claims based on detecting broken phones on images provided by customers.
  Enviropment: Python, Yolo, Mediapipe, SAM, OpenCV, GIT

- **Computer Vision for Robots | Memristor | Computer Vision Team Lead | Team size: 3**
  Responsibilities: Creating systems for object detection, classification and localization. Making datasets, processing data, training models and using in real systems. Creating frontent for the poc and mvp.
  Enviropment: Python, React, YOLO, OpenCV, Raspberry pi, GIT

- **Software for Autonomous Robots Movements | Memristor | Python Developer|Team size: 6**
  Responsibilities: Programming robots for autonomous movements. Leading and organizing team.
  Enviropment: Python, Raspberry pi, GIT

- **ML Platform Migration to Ray| Freelance | ML engineer | 3**

  Responsibilities: Migrating ML training workflows from local environments to distributed Ray platform. Refactoring training pipelines to integrate Ray, AWS S3, and Comet, replacing Torch-based local execution, NAS storage, and MLflow. Reorganizing codebase and migrating datasets to AWS S3 and FSx for scalable storage and distributed training.

  Enviropment: Python, Ray, AWS S3, FSx, Comet ML, PyTorch

- **Django – React data platform | Freelance | Team Lead and AI Specialist | 6**

  Responsibilities: Leading full stack team. Comunicating with client, organizing team, developing Django funcionalities, AWS setup. Creating system for extracting data from unstructured textual documents and Apollo findings.

  Enviropment: Python, Django, AWS, Github

Additional Interests

- **Photography**
- **Travel**
- **Road Trips**
- **Camping**
- **Cycling**
- **Hobby programming -** more about projects on www.dusanblanusa.com